DATA 621 Blog 4

Regression with Interactions

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Interaction

Regression is a statistical method used to model and predict an outcome variable based on one or more explanatory variables. In linear regression, the relationship is assumed to be linear and can be expressed as the value of the response variable y as a function of a predictor x and an intercept. Formally, this can be written:

$$y = \beta 0 + \beta 1x + \varepsilon$$

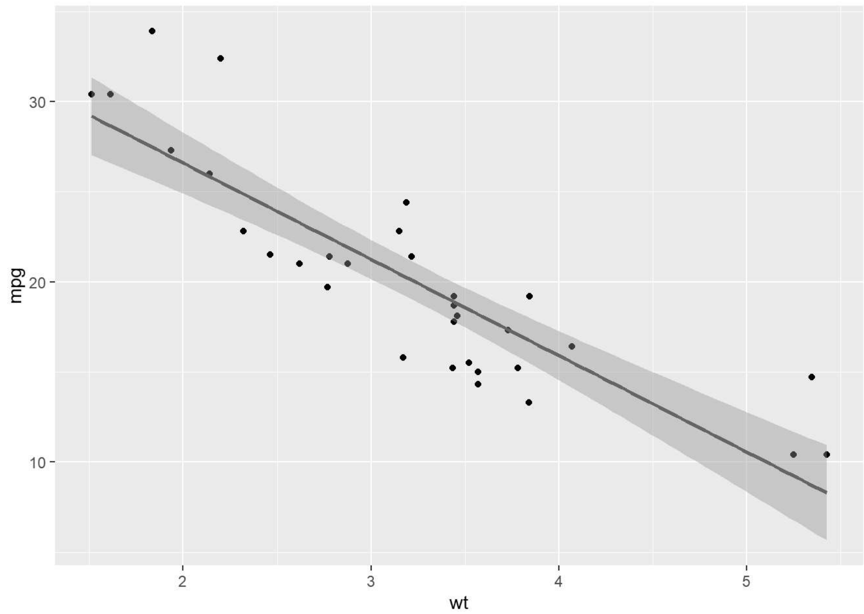
where β0 is the intercept, β1 represents the effect of x on y, and ε captures random error.

While multiple predictors can be included in a regression model, visualization in two dimensions requires holding other variables constant or grouping by categorical predictors. Interaction plots allow us to visually assess whether the relationship between a predictor and the outcome differs across levels of another variable.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 \times x_2) + \epsilon$$

▶ Code

```
`geom_smooth()` using formula = 'y ~ x'
```



Here, our regression model predicts miles per gallon (mpg) as a function of vehicle weight (wt). The fitted line indicates that as vehicle weight increases, the expected fuel efficiency decreases.

▶ Code

```
Call:
glm(formula = mtcars$mpg ~ mtcars$wt, data = mtcars)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
mtcars$wt    -5.3445     0.5591  -9.559 1.29e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 9.277398)

    Null deviance: 1126.05  on 31  degrees of freedom
Residual deviance:  278.32  on 30  degrees of freedom
AIC: 166.03

Number of Fisher Scoring iterations: 2
```

Intuitively, vehicles with a greater number of cylinders tend to have higher energy demands. Allowing cylinder count to